

Gor'kov Potentials and Acoustic Levitation: A Self-Contained Treatment

From the wave equation to trapping forces, with worked problems

Companion notes to the DeerFlow acoustic-levitator simulation

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Abstract

These notes develop, from first principles, everything needed to understand the **Gor'kov potential** — the scalar function whose negative gradient gives the time-averaged acoustic radiation force on a small particle. We cover the linearised equations of acoustics, the harmonic (complex) representation of sound fields, the acoustic radiation-stress tensor, Rayleigh scattering by a small compressible sphere, the derivation and validity of the Gor'kov potential, trap stability and stiffness, and the extension to multi-transducer phased arrays (the geometry of a real acoustic levitator). A set of practice problems with full solutions closes each section. The treatment is aimed at readers who want to be able to *compute*, not merely quote, the trapping behaviour of a levitator such as a 72-transducer, 40 kHz ultrasonic array.

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1 Linear acoustics in one page

Acoustic levitation operates in the regime of *small perturbations* about a quiescent reference state. We denote the equilibrium density, pressure, and speed of sound of the host fluid by ρ_0 , p_0 , and c_0 . The fields are written as

$$\rho(\mathbf{x}, t) = \rho_0 + \rho_1(\mathbf{x}, t), \quad p(\mathbf{x}, t) = p_0 + p_1(\mathbf{x}, t), \quad \mathbf{v}(\mathbf{x}, t) = \mathbf{v}_1(\mathbf{x}, t), \quad (1)$$

where all subscript-1 quantities are first-order small.

The three equations. Conservation of mass, conservation of momentum (Euler's equation, viscosity neglected), and the isentropic equation of state give, to first order,

$$\frac{\partial \rho_1}{\partial t} + \rho_0 \nabla \cdot \mathbf{v}_1 = 0, \quad (2a)$$

$$\rho_0 \frac{\partial \mathbf{v}_1}{\partial t} = -\nabla p_1, \quad (2b)$$

$$p_1 = c_0^2 \rho_1, \quad (2c)$$

with

$$c_0^2 = \left(\frac{\partial p}{\partial \rho} \right)_s = \frac{1}{\rho_0 \kappa_0}, \quad (3)$$

where κ_0 is the *adiabatic compressibility* $\kappa_0 \equiv -(1/V)(\partial V/\partial p)_s$. For an ideal gas, $c_0^2 = \gamma p_0/\rho_0$ with γ the ratio of specific heats; for air at 20 °C, $c_0 \approx 343$ m/s.

The wave equation. Taking the divergence of (2b), substituting (2a) and (2c) eliminates ρ_1 and $\mathbf{v}_1$:

$$\nabla^2 p_1 - \frac{1}{c_0^2} \frac{\partial^2 p_1}{\partial t^2} = 0. \quad (4)$$

1.1 Harmonic fields and complex notation

It is extremely convenient to work with time-harmonic fields using the complex convention

$$p_1(\mathbf{x}, t) = \text{Re}[P(\mathbf{x}) e^{-i\omega t}], \quad \mathbf{v}_1(\mathbf{x}, t) = \text{Re}[\mathbf{u}(\mathbf{x}) e^{-i\omega t}], \quad (5)$$

where P and $\mathbf{u}$ are complex amplitudes. Substituting into (4) gives the **Helmholtz equation**

$$\nabla^2 P + k^2 P = 0, \quad k = \frac{\omega}{c_0} = \frac{2\pi}{\lambda}, \quad (6)$$

and (2b) relates velocity to pressure through

$$\mathbf{u} = -\frac{1}{i\omega\rho_0} \nabla P = \frac{i}{\omega\rho_0} \nabla P. \quad (7)$$

Two observations about (7) are used constantly below: (i) velocity is *in quadrature* with the pressure gradient (the factor i), and (ii) where P has an extremum, $\mathbf{u} = 0$, and vice versa.

Time averages. The time average of a product of two real harmonic quantities $A = \text{Re}[a e^{-i\omega t}]$, $B = \text{Re}[b e^{-i\omega t}]$ over one period is

$$\langle AB \rangle = \frac{1}{T} \int_0^T AB \, dt = \frac{1}{2} \text{Re}[a b^*] = \frac{1}{2} \text{Re}[a^* b]. \quad (8)$$

In particular,

$$\langle p^2 \rangle = \frac{1}{2} |P|^2, \quad \langle v^2 \rangle = \frac{1}{2} |\mathbf{u}|^2 = \frac{1}{2\omega^2 \rho_0^2} |\nabla P|^2. \quad (9)$$

These two quadratic averages, $\langle p^2 \rangle$ and $\langle v^2 \rangle$, are the *only* field quantities that enter the Gor'kov potential.

Two canonical fields. A *plane travelling wave* $P = P_0 e^{ikz}$ has $\langle p^2 \rangle = P_0^2/2$ and $\langle v^2 \rangle = P_0^2/(2\rho_0^2 c_0^2)$, both uniform. A *standing wave* formed by counter-propagating waves, $P = 2P_0 \cos(kz)$, has

$$\langle p^2 \rangle = 2P_0^2 \cos^2(kz), \quad \langle v^2 \rangle = \frac{2P_0^2}{\rho_0^2 c_0^2} \sin^2(kz). \quad (10)$$

The *pressure nodes* (where $P = 0$ and $\langle p^2 \rangle = 0$) are at $kz = \pi/2 + n\pi$, i.e. $z = \lambda/4 + n\lambda/2$; the *pressure antinodes* are at $z = n\lambda/2$. This node/antinode structure is the backbone of standing wave levitation.

1.2 Numerical anchors for a 40 kHz levitator

Quantity	Symbol	Value (air, 20 °C)
speed of sound	c_0	343 m/s
density	ρ_0	1.2 kg/m ³
adiabatic compressibility	$\kappa_0 = 1/(\rho_0 c_0^2)$	$\approx 7.1 \times 10^{-6} \text{ Pa}^{-1}$
frequency	f	40 kHz
angular frequency	$\omega = 2\pi f$	$2.51 \times 10^5 \text{ rad/s}$
wavelength	$\lambda = c_0/f$	$\approx 8.58 \text{ mm}$
wavenumber	$k = 2\pi/\lambda$	$\approx 732 \text{ m}^{-1}$

(Transducer modules of the levitator's type are specified at $\lambda = 8.57 \text{ mm}$, essentially the same value.) The half-wavelength spacing of standing-wave nodes is $\lambda/2 \approx 4.29 \text{ mm}$, which is why levitators are built with their opposing transducer arrays roughly a few half-wavelengths apart.

2 Acoustic radiation force and the radiation-stress tensor

Acoustic levitation is a *second-order* effect: the force is quadratic in the (linear) sound field. Over one cycle a particle feels no net first-order force from a harmonic wave, but the nonlinear advection and equation-of-state terms produce a small steady *radiation pressure* that does not average to zero.

2.1 Momentum flux and radiation stress

Momentum conservation for the fluid implies that the force exerted by a sound field on a body enclosed by a surface S (with outward normal $\mathbf{n}$) is the rate at which momentum flows out of S ,

$$\mathbf{F} = - \oint_S \langle \mathbf{\Pi} \rangle \cdot \mathbf{n} \, dS, \quad (11)$$

where $\mathbf{\Pi}$ is the momentum-flux tensor. To second order in the acoustic field, the time-averaged **radiation-stress tensor** for a non-viscous fluid is

$$\langle \Pi_{ij} \rangle = \rho_0 \langle v_i v_j \rangle + \left[\frac{\langle p^2 \rangle}{2\rho_0 c_0^2} - \frac{\rho_0}{2} \langle v^2 \rangle \right] \delta_{ij}. \quad (12)$$

The first term is the momentum carried by the oscillatory flow; the bracket is a small isotropic “pressure” arising from the nonlinear equation of state and from the second-order advection.

Remark 2.1 (Eulerian vs. Lagrangian radiation pressure). (12) is the Eulerian momentum flux (fixed in space). A second, uniform term $\langle p^2 \rangle \delta_{ij}$ involving the second-order mean pressure appears when one works with Lagrangian (material) coordinates. For the net force on a particle this uniform term integrates to zero around any closed surface, so it does not enter the Gor’kov potential. It does matter for the radiation pressure measured on a piston or wall.

Example 2.1 (Radiation pressure of a plane wave). For a plane travelling wave $P = P_0 e^{ikz}$, $\langle p^2 \rangle = P_0^2/2$ and $\langle v^2 \rangle = P_0^2/(2\rho_0^2 c_0^2)$, so the bracket in (12) vanishes and $\langle \Pi_{zz} \rangle = \rho_0 \langle v_z^2 \rangle = P_0^2/(2\rho_0 c_0^2)$, the well-known radiation pressure on a perfect absorber. On a perfect reflector the pressure doubles to $P_0^2/(\rho_0 c_0^2)$.

2.2 The Gor’kov strategy

Evaluating (11) for a body of *arbitrary* size requires solving the full scattering problem and integrating. Gor’kov’s key simplification (1962) applies to a *small* particle, $ka \ll 1$. In that limit the particle scatters as a **monopole** plus a **dipole** only, the scattered field is known in closed form, and — remarkably — the resulting force can be written as the negative gradient of a scalar potential. That potential is the Gor’kov potential.

3 Scattering by a small compressible sphere

Consider a sphere of radius a , density ρ_p , and compressibility κ_p , immersed in a fluid (ρ_0, κ_0, c_0) , small enough that $ka \ll 1$ so that the incident field is locally uniform across the particle. Two distinct physical mechanisms scatter sound.

3.1 Monopole scattering: compressibility contrast

A pressure fluctuation p_1 compresses the particle by a relative volume change $\Delta V/V = -\kappa_p p_1$, whereas the surrounding fluid would compress by $-\kappa_0 p_1$. The *difference* in compressibility acts as a source of volume, i.e. a monopole. The volume-velocity source strength is

$$q(t) = \frac{\partial}{\partial t} [(\kappa_p - \kappa_0) V_p p_1] = (\kappa_p - \kappa_0) V_p \frac{\partial p_1}{\partial t}, \quad (13)$$

and a monopole of strength q radiates $p_{\text{mono}} = \frac{\rho_0}{4\pi r} \frac{\partial q}{\partial t}$. Using $\partial^2 p_1 / \partial t^2 = -\omega^2 p_1$ and $\rho_0 \omega^2 \kappa_0 = k^2$,

$$p_{\text{mono}}(\mathbf{r}) = - \left(1 - \frac{\kappa_p}{\kappa_0} \right) \frac{k^2 V_p}{4\pi r} p_{\text{in}} = -f_1 \frac{k^2 V_p}{4\pi r} p_{\text{in}}, \quad (14)$$

where we define the **monopole factor**

$$f_1 = 1 - \frac{\kappa_p}{\kappa_0} = 1 - \frac{\rho_0 c_0^2}{\rho_p c_p^2} \quad (15)$$

and c_p is the speed of sound inside the particle. This is exactly the result obtained by the full Rayleigh-scattering analysis.

3.2 Dipole scattering: density contrast

A particle denser (or lighter) than the fluid does not follow the fluid's oscillation perfectly. In a uniform oscillating flow $\mathbf{u}_{\text{in}}$, a rigid sphere with added mass $\frac{1}{2}\rho_0 V_p$ oscillates with velocity

$$\mathbf{u}_p = \frac{3\rho_0}{2\rho_p + \rho_0} \mathbf{u}_{\text{in}}, \quad (16)$$

so the particle *slips* relative to the fluid by

$$\mathbf{u}_p - \mathbf{u}_{\text{in}} = \left(\frac{3\rho_0}{2\rho_p + \rho_0} - 1 \right) \mathbf{u}_{\text{in}} = - \underbrace{\frac{2(\rho_p - \rho_0)}{2\rho_p + \rho_0}}_{f_2} \mathbf{u}_{\text{in}} = -f_2 \mathbf{u}_{\text{in}}. \quad (17)$$

A sphere translating with velocity $\mathbf{u}$ through a fluid produces the potential flow $\phi = -\frac{1}{2}a^3(\mathbf{u} \cdot \nabla)(1/r)$; replacing $\mathbf{u}$ by the slip velocity (17) and using $p = -\rho_0 \partial\phi/\partial t$ gives the **dipole scattered pressure**

$$p_{\text{dip}}(\mathbf{r}) = -\frac{\rho_0 a^3}{2} f_2 \frac{\partial \mathbf{u}_{\text{in}}}{\partial t} \cdot \nabla \left(\frac{1}{r} \right), \quad (18)$$

with the **dipole factor**

$$f_2 = \frac{2(\rho_p - \rho_0)}{2\rho_p + \rho_0} \quad (19)$$

(the factor f_2 is precisely the normalized density contrast appearing in the classical result for an oscillating sphere).

Remark 3.1 (Signs and limits). $f_2 \rightarrow 1$ for a rigid, immobile particle ($\rho_p \rightarrow \infty$) and $f_2 \rightarrow -2$ for a massless particle ($\rho_p \rightarrow 0$, e.g. a gas bubble in a liquid). The monopole factor $f_1 \rightarrow 1$ for an incompressible particle ($\kappa_p \rightarrow 0$) and $f_1 \rightarrow -\infty$ for a highly compressible particle such as a bubble in water.

4 Derivation of the Gor'kov potential

4.1 Setup

The total field is the sum of the incident field ($p_{\text{in}}, \mathbf{u}_{\text{in}}$) and the scattered field ($p_s, \mathbf{u}_s$). Substituting $p = p_{\text{in}} + p_s$, $\mathbf{v} = \mathbf{u}_{\text{in}} + \mathbf{u}_s$ into (12) and integrating over a surface S enclosing the particle,

$$\mathbf{F} = - \oint_S \langle \mathbf{\Pi} \rangle \cdot \mathbf{n} \, dS, \quad (20)$$

one finds that the integrand splits into terms involving only the incident field, only the scattered field, and cross terms. The pure-incident terms integrate to zero around a closed surface (no force from a field in the absence of a particle); the pure-scattered terms also vanish to leading order for $ka \ll 1$. The force therefore comes from the *cross terms* between the incident and scattered fields. Using (14) and (18) and carrying out the surface integral (which the reader is invited

to perform in Problem 6.5; the monopole term uses $\oint_S r^{-1} dS = 4\pi a$ and the dipole term uses $\oint_S r_i r_j / r^2 dS = \frac{4\pi a^2}{3} \delta_{ij}$, one obtains

$$\mathbf{F} = V_p \nabla \left[f_1 \frac{\langle p^2 \rangle}{2\rho_0 c_0^2} - f_2 \frac{3\rho_0}{4} \langle v^2 \rangle \right], \quad (21)$$

with $V_p = \frac{4}{3}\pi a^3$ the particle volume.

4.2 The potential

Because (21) is a pure gradient, we may write $\mathbf{F} = -\nabla U$ with the **Gor'kov potential**

$$U(\mathbf{x}) = V_p \left[f_1 \frac{\langle p^2 \rangle}{2\rho_0 c_0^2} - f_2 \frac{3\rho_0}{4} \langle v^2 \rangle \right], \quad (22)$$

where $\langle p^2 \rangle = \frac{1}{2}|P|^2$ and $\langle v^2 \rangle = \frac{1}{2}|\mathbf{u}|^2$ are evaluated at the (unperturbed) particle position. Equation (22) is the central result of these notes. Two physical statements are packed into it:

- (i) **Pressure term.** The positive term $f_1 \langle p^2 \rangle / (2\rho_0 c_0^2)$ is the monopole contribution. For $f_1 > 0$ it *repels* the particle from pressure maxima; the energy is minimized where $\langle p^2 \rangle$ is minimized (pressure nodes).
- (ii) **Velocity term.** The negative term $-f_2(3\rho_0/4) \langle v^2 \rangle$ is the dipole contribution. For $f_2 > 0$ it *attracts* the particle toward velocity maxima. Since velocity maxima coincide with pressure nodes, the two terms *reinforce* each other for a dense, stiff particle in a gas.

The force is

$$\mathbf{F} = -\nabla U, \quad (23)$$

and in the absence of gravity a particle released in the field slides downhill in U and settles at a local minimum of U — a *trap*.

4.3 Physical meaning: nodes vs. antinodes

The sign of f_1 and f_2 decides whether a particle is trapped at a pressure *node* or a pressure *antinode*. The four cases of practical interest are:

Particle / host	f_1	f_2	trapped at	Example
solid in gas	> 0	> 0	pressure node	polystyrene in air
liquid droplet in gas	> 0	> 0	pressure node	water in air
solid in liquid	$\lesssim 0$	> 0	near node (balance)	cells in water
gas bubble in liquid	< 0	< 0	pressure antinode	air in water

For the levitator of interest, the particles are solids or droplets in *air*. Air is extraordinarily compressible and light compared to any solid or liquid: $\kappa_p \ll \kappa_0$ and $\rho_p \gg \rho_0$. Hence

$$f_1 \approx 1, \quad f_2 \approx 1, \quad (\text{solid/droplet in air}) \quad (24)$$

and the potential simplifies to

$$U \approx V_p \left[\frac{\langle p^2 \rangle}{2\rho_0 c_0^2} - \frac{3\rho_0}{4} \langle v^2 \rangle \right]. \quad (25)$$

With $f_1 = f_2 = 1$ the particle is trapped at a pressure node, where both $\langle p^2 \rangle$ and the repulsive term vanish and the attractive velocity term is maximized. This is why ultrasonic levitators hold particles at the nodes of the standing wave.

4.4 Validity conditions

The Gor'kov potential is reliable when all of the following hold:

- (a) **Rayleigh limit** $ka \ll 1$ (particle much smaller than the wavelength). For 40 kHz in air, $ka \lesssim 0.46$ for $a \lesssim 1$ mm, i.e. spheres up to ~ 2 mm diameter are reasonably handled; larger particles need the full Mie-type series.
- (b) **Weak scattering**: the field at the particle position is the incident field (single scattering, no re-scattering between particles).
- (c) **Non-viscous host** and negligible acoustic streaming relative to the radiation force. (Streaming is a separate, typically weaker steady flow.)
- (d) **Particle treated as a small sphere**; non-spherical particles need a multipole generalization (the same idea with more terms).

Example 4.1 (Trap depth for a rigid sphere in a standing wave). *Using (10) in (25) gives $U(z) = V_p P_0^2 / (\rho_0 c_0^2) [\cos^2(kz) - \frac{3}{4} \sin^2(kz)]$ (see Problem 6.2). At a node ($kz = \pi/2$) the value is $U = -\frac{3}{4} V_p P_0^2 / (\rho_0 c_0^2)$, while at an antinode ($kz = 0$) it is $U = +V_p P_0^2 / (\rho_0 c_0^2)$. The trap depth (energy difference between antinode and node) is therefore $\Delta U = \frac{7}{4} V_p P_0^2 / (\rho_0 c_0^2)$.*

5 From one transducer to 72: phased arrays and holography

A single transducer produces a beam; two opposing transducers produce a standing wave with a single nodal plane. A *phased array* — many transducers whose individual phases (and, optionally, amplitudes) are controlled independently — can synthesize a nearly arbitrary acoustic field in a volume, and therefore a nearly arbitrary arrangement of traps. This is the principle behind the 72-transducer levitator studied here.

5.1 Superposition of piston fields

Label the transducers $n = 1, \dots, N$ (here $N = 72$), each located at $\mathbf{x}_n$ with axis direction $\hat{\mathbf{b}}_n$, radius R , phase φ_n , and amplitude weight w_n . The total complex pressure is the coherent sum

$$P(\mathbf{x}) = \sum_{n=1}^N w_n D_n(\mathbf{x}) P_0 e^{i(kd_n + \varphi_n)}, \quad (26)$$

where $d_n = |\mathbf{x} - \mathbf{x}_n|$ and D_n is the single-transducer directivity pattern (Section 5.2). The particle velocity follows from (2b),

$$\mathbf{u}(\mathbf{x}) = \frac{i}{\rho_0 \omega} \nabla P(\mathbf{x}), \quad (27)$$

and both are substituted into the Gor'kov potential (22).

5.2 Baffled-piston directivity

Each 40 kHz transducer is a disc of radius $R \approx 5$ mm mounted in a baffle. A rigid circular piston of radius R vibrating with normal velocity v_0 in an infinite rigid baffle radiates the *Rayleigh–Sommerfeld* field

$$P_{\text{piston}}(r, \theta) = \frac{\rho_0 c v_0 k}{2\pi} \int_0^R \int_0^{2\pi} \frac{e^{ik|\mathbf{r}-\mathbf{r}'|}}{|\mathbf{r}-\mathbf{r}'|} r' dr' d\phi', \quad (28)$$

which in the far field ($r \gg R$, $r \gg kR^2$) reduces to

$$P(r, \theta) = i \rho_0 c v_0 \frac{R^2}{r} e^{ikr} \underbrace{\frac{J_1(kR \sin \theta)}{kR \sin \theta}}_{D(\theta)}, \quad (29)$$

where J_1 is the Bessel function of the first kind and $D(\theta)$ is the **directivity**. The directivity has a main lobe on axis ($\theta = 0$, $D = 1$) and nulls at $kR \sin \theta = \gamma_{1m}$ (zeros of J_1); the first null is at $\sin \theta = \gamma_{11}/(kR) \approx 3.83/(kR)$. Because $kR \approx 1.2$ at 40 kHz, the pattern is broad (near-hemispherical), which is important: each transducer contributes over a wide solid angle and many transducers cooperate to form a single trap.

In the near field the exact (28) is used; the simulation evaluates it directly so that traps a fraction of a wavelength from the array are represented correctly. For axisymmetric computation the double integral can be collapsed to a one-dimensional quadrature, which is what the code does.

5.3 Phase-only holography for trap shaping

Given a desired pressure target $P_{\text{target}}(\mathbf{x})$ (e.g. bright spots at chosen trap positions), one may solve for the phases φ_n that best reproduce it. Because amplitude control adds cost and complexity, the practical scheme is **phase-only** control ($w_n = 1$), which relies on the fact that the *phase* of each transducer carries most of the information needed to form traps. A standard choice is the *backpropagation* (time-reversal) prescription: to focus energy at $\mathbf{x}_t$, set

$$\varphi_n = -k |\mathbf{x}_t - \mathbf{x}_n| = -kd_{n,t}, \quad (30)$$

so that all contributions arrive *in phase* at $\mathbf{x}_t$ and interfere constructively. This is the exact analogue of time-reversal focusing and of a phased-array “delay-and-sum” beamformer. It reproduces the acoustic *hologram* used in modern levitators.

Remark 5.1 (Hologram vs. potential). *Focusing produces a pressure maximum. For particles with $f_1 > 0$ the trap is a pressure node, so the target U -minimum is not a pressure maximum but a null flanked by lobes. In practice one either (i) focuses a ring of lobes whose center is a node, or (ii) targets a local minimum of the full U directly by optimizing φ_n against U rather than against $|P|$. The latter is what the levitator code does for trap placement.*

5.4 Standing waves, focusing, steering, and vortices

By choosing the phases, one synthesizes the canonical trap families:

Standing wave. φ_n constant, equal phase on both shells of the opposed array $\Rightarrow$ axial standing wave with nodal planes spaced $\lambda/2$ apart (Section ??).

Focus / single trap. Backpropagation (30) to one point concentrates the field and creates a tightly localized potential well.

Steering / translation. Adding a linear phase ramp $\varphi_n = \varphi_n^0 - k \hat{\mathbf{s}} \cdot \mathbf{x}_n$ translates the trap along $\hat{\mathbf{s}}$ (a phased-array beamsteer).

Vortex trap (topological charge m). The phase $\varphi_n = m \arg(\mathbf{x}_t - \mathbf{x}_n)_\perp$ (azimuthal phase winding by $2\pi m$ around the axis) creates an acoustic vortex — a field with a *phase singularity* on the axis. The central node is an ideal trap for a particle or droplet, and the field carries orbital angular momentum.

Twin / multiple traps. Summing backpropagation phases for several target points (or partitioning the array into sub-arrays) produces several traps simultaneously; this is used for multi-particle and contactless assembly.

5.5 Orbital angular momentum and rotation

A vortex field $P(r, \phi, z) \propto A(r, z) e^{im\phi}$ carries a phase singularity at $r = 0$ where $A \rightarrow 0$, hence a deep pressure node. The azimuthal phase gradient $\frac{1}{r} \frac{\partial}{\partial \phi}(m\phi) = m/r$ corresponds to an azimuthal velocity, and the field transports **orbital angular momentum** (OAM) along the axis. A particle trapped on (or slightly off) the vortex core experiences an azimuthal radiation *torque* and rotates (Problem 6.8). Varying m and the phase offsets lets one spin, translate, and manipulate particles, droplets, and even small objects with no contact.

Example 5.1 (72-transducer geometry). *The TinyLev-style array in the STL file has two opposed spherical caps, each with 36 transducers arranged in concentric rings of 6, 12, and 18 elements, on a sphere of radius $R_c = 90$ mm aimed at the center. The operating frequency $f = 40$ kHz gives $\lambda = c/f \approx 8.65$ mm in air, so a levitated droplet of ~ 2 mm diameter satisfies $ka \approx 0.46 \lesssim 1$ and the Gor'kov picture is valid. The opposed geometry forms a standing wave in the center; adding holographic phases turns that standing wave into steerable single or multiple traps.*

6 Practice problems

Problems progress from basic evaluation to full derivations. Full solutions to the guided problems follow at the end of the section.

Problem 6.1 (Monopole and dipole factors). *Compute f_1 and f_2 for (a) a polystyrene sphere ($\rho_p = 1050$ kg/m³, $c_p = 2350$ m/s) in air ($\rho_0 = 1.2$ kg/m³, $c = 343$ m/s); (b) an air bubble in water ($\rho_p = 1.2$, $c_p = 343$, $\rho_0 = 1000$, $c = 1480$). In each case state whether the particle is trapped at a pressure node or antinode.*

Problem 6.2 (Standing-wave potential). *A plane standing wave has $P(z) = P_0 \cos(kz)$ with P_0 real. Using $\mathbf{u} = \frac{i}{\rho_0 \omega} \nabla P$, show that $\langle p^2 \rangle = \frac{1}{2} P_0^2 \cos^2(kz)$ and $\langle v^2 \rangle = \frac{P_0^2}{2\rho_0^2 c^2} \sin^2(kz)$. Hence derive*

$$U(z) = \frac{V_p P_0^2}{\rho_0 c^2} \left[f_1 \cos^2(kz) - \frac{3}{4} f_2 \sin^2(kz) \right]. \quad (31)$$

For $f_1 = f_2 = 1$, locate the potential minima and confirm they are at $kz = \frac{\pi}{2}, \frac{3\pi}{2}, \dots$ (pressure nodes), spaced $\lambda/2$ apart.

Problem 6.3 (Trap depth). *For $f_1 = f_2 = 1$, compute the trap depth ΔU between a node and an adjacent antinode in (31). Evaluate it in joules for a water droplet of radius $a = 0.5$ mm in air at $P_0 = 1.5$ kPa. Compare with the gravitational energy $mg \lambda/2$ of the droplet. What does the comparison imply about the minimum acoustic pressure needed to levitate it?*

Problem 6.4 (Pressure vs. velocity contrast). A particle has f_1 and f_2 both positive. Explain, from (22), why the pressure term repels the particle from pressure antinodes while the velocity term attracts it to velocity antinodes, and why these two statements are consistent (recall the phase relation between p and v in a standing wave).

Problem 6.5 (Deriving the force from momentum flux). This is the derivation skipped in Section 4. Consider a rigid sphere of radius a in a fluid with an incident plane wave.

- (a) Write the total pressure as $p = p_{\text{in}} + p_s$ and velocity $\mathbf{v} = \mathbf{u}_{\text{in}} + \mathbf{u}_s$. Expand the time-averaged stress (12) to first order in the scattered field and identify the cross terms.
- (b) Show that the pure-incident terms integrate to zero over a sphere enclosing the particle, so the force is $\mathbf{F} = - \oint_S \langle \mathbf{\Pi} \rangle_{\text{cross}} \cdot \mathbf{n} \, dS$.
- (c) Using the monopole field (14), show the monopole contribution is $\mathbf{F}_{\text{mono}} = V_p f_1 \nabla (\langle p^2 \rangle / (2\rho_0 c^2))$, using $\oint_S r^{-1} \, dS = 4\pi a$ and $\nabla \langle p^2 \rangle$ evaluated at the center.
- (d) Using the dipole field (18), show the dipole contribution is $\mathbf{F}_{\text{dip}} = -V_p f_2 \frac{3\rho_0}{4} \nabla \langle v^2 \rangle$, using $\oint_S r_i r_j / r^2 \, dS = \frac{4\pi a^2}{3} \delta_{ij}$.
- (e) Combine (c) and (d) to recover (21) and the Gor'kov potential (22).

Problem 6.6 (Rayleigh limit of the monopole). A compressible sphere of radius a sits in a plane wave $p_{\text{in}} = P_0 e^{ikz}$. In the long-wavelength limit $ka \ll 1$, the exact Mie-type series for a compressible sphere reduces to

$$p_s(\mathbf{r}) = -\frac{k^2 V_p}{4\pi r} f_1 P_0 - \frac{\rho_0 a^3}{2} f_2 \frac{\partial u_{\text{in}}}{\partial t} \cdot \nabla \left(\frac{1}{r} \right). \quad (32)$$

Identify the monopole and dipole terms and confirm they match (14) and (18). Explain why higher multipoles (quadrupole, etc.) are suppressed by powers of $(ka)^2$.

Problem 6.7 (Trapping criterion and stability). For a particle with $f_1, f_2 > 0$ in the standing wave (31), compute the force $F(z) = -dU/dz$ and the trap stiffness $k_t = d^2U/dz^2$ at a node. Show that $k_t > 0$ (stable) requires $f_1 + \frac{3}{4}f_2 > 0$. Interpret this condition physically and check it for polystyrene in air.

Problem 6.8 (Radiation torque and OAM). A vortex trap has $P(r, \phi, z) = A(r, z) e^{im\phi}$ with $A \rightarrow 0$ as $r \rightarrow 0$.

- (a) Show the azimuthal phase gradient is m/r and hence the field carries angular momentum density proportional to $m |A|^2$ (time-averaged).
- (b) The azimuthal component of the radiation force on an off-axis particle produces a torque $\boldsymbol{\tau} = \mathbf{r} \times \mathbf{F}$. Argue that a particle slightly displaced from the core experiences a torque proportional to m , and hence rotates.
- (c) For the phase assignment $\varphi_n = m \arg(\mathbf{x}_t - \mathbf{x}_n)_\perp$ on the 72-transducer array, explain how the superposition (26) produces a field approximating $e^{im\phi}$ near the axis.

Problem 6.9 (Phase-only holography). You want a single trap at $\mathbf{x}_t$ using backpropagation $\varphi_n = -k|\mathbf{x}_t - \mathbf{x}_n|$ (30).

- (a) Show that all 72 contributions arrive in phase at $\mathbf{x}_t$, so $|P(\mathbf{x}_t)|$ is maximized.

- (b) Explain why, for $f_1 > 0$, the trap is nevertheless a pressure node, and why the simple back-propagation focus is not by itself a Gor'kov minimum.
- (c) Propose a target $U(\mathbf{x})$ (rather than $|P|$) and describe an iterative scheme $\varphi_n \leftarrow \arg \min_{\varphi} U(\mathbf{x}_t)$ (e.g. a few steps of a projected gradient on the phases) that converges to a true potential well.

Problem 6.10 (Scaling with frequency). The Gor'kov potential scales as $U \propto P_0^2/(\rho_0 c^2)$ at fixed field amplitude, but a transducer of fixed power produces a pressure that scales with k (via the piston far field (29)). Argue that, for fixed acoustic power, U scales with ω^2 , so higher frequency traps more strongly — but at the cost of a shorter wavelength λ and hence a smaller maximum levitable particle. Discuss the trade-off for 40 kHz vs. 100 kHz levitators.

Problem 6.11 (Design: levitating a water drop). A 72-transducer, 40 kHz levitator holds a 0.5 mm-radius water droplet ($\rho_p = 1000 \text{ kg/m}^3$) in air.

- (a) Verify $ka < 1$ and justify the Gor'kov approximation.
- (b) Compute f_1, f_2 to confirm $f_1 \approx f_2 \approx 1$.
- (c) Given $P_0 = 1.5 \text{ kPa}$, compute the trap depth and the trap stiffness, and estimate the restoring force for a $50 \mu\text{m}$ displacement from the node.
- (d) The droplet weight is mg . What minimum pressure amplitude is needed for the trap to support the droplet? Does it exceed the value in (c)?

Solutions and guided steps

6.1. (a) $f_1 = 1 - \frac{\rho_0 c_p^2}{\rho_p c_p^2} = 1 - \frac{1.2 \cdot 343^2}{1050 \cdot 2350^2} \approx 0.99998$, $f_2 = \frac{2(\rho_p - \rho_0)}{2\rho_p + \rho_0} \approx 1.0$. Both positive $\Rightarrow$ trapped at a pressure *node*. (b) For a bubble: $f_1 = 1 - \frac{1000 \cdot 1480^2}{1.2 \cdot 343^2} \approx -1.6 \times 10^4$, $f_2 = \frac{2(1.2 - 1000)}{2 \cdot 1.2 + 1000} \approx -1.99$. Both negative $\Rightarrow$ trapped at a pressure *antinode* (classic result: bubbles migrate to pressure maxima).

6.2. With $P = P_0 \cos kz$ real, $u = \frac{i}{\rho_0 \omega} \frac{dP}{dz} = \frac{-i P_0 k}{\rho_0 \omega} \sin kz = -i \frac{P_0}{\rho_0 c} \sin kz$. Then $\langle p^2 \rangle = \frac{1}{2} |P|^2 = \frac{1}{2} P_0^2 \cos^2 kz$ and $\langle v^2 \rangle = \frac{1}{2} |u|^2 = \frac{P_0^2}{2\rho_0^2 c^2} \sin^2 kz$. Substitution into (22) gives (31). For $f_1 = f_2 = 1$, $U(z) = \frac{V_p P_0^2}{\rho_0 c^2} [\cos^2 kz - \frac{3}{4} \sin^2 kz]$. Differentiating: $\frac{dU}{dz} = -\frac{V_p P_0^2 k}{\rho_0 c^2} \cdot \frac{7}{2} \sin kz \cos kz$, zero where $\sin kz = 0$ (antinodes, $U > 0$) or $\cos kz = 0$ (nodes, $U < 0$). The minima are at $\cos kz = 0$, i.e. $kz = \pi/2, 3\pi/2, \dots$

6.3. At a node $U_- = -\frac{3}{4} \frac{V_p P_0^2}{\rho_0 c^2}$; at an antinode $U_+ = +\frac{V_p P_0^2}{\rho_0 c^2}$; so $\Delta U = \frac{7}{4} \frac{V_p P_0^2}{\rho_0 c^2}$. With $V_p = \frac{4}{3} \pi (0.5 \text{ mm})^3 = 5.24 \times 10^{-10} \text{ m}^3$, $\rho_0 c^2 = 1.2 \cdot 343^2 \approx 1.41 \times 10^5 \text{ Pa}$, $P_0^2 = 2.25 \times 10^6 \text{ Pa}^2$: $\Delta U = \frac{7}{4} \frac{5.24 \times 10^{-10} \cdot 2.25 \times 10^6}{1.41 \times 10^5} \approx 5.9 \times 10^{-8} \text{ J}$. Gravity over $\lambda/2 = 4.3 \text{ mm}$: $mg \lambda/2 = \rho_p V_p g \lambda/2 = 1000 \cdot 5.24 \times 10^{-10} \cdot 9.8 \cdot 4.3 \times 10^{-3} \approx 2.2 \times 10^{-8} \text{ J}$. Since $\Delta U > mg \lambda/2$, the trap can support the droplet at this pressure; levitation requires $P_0 \gtrsim \sqrt{\frac{4}{7} \rho_0 c^2 \rho_p g \lambda/2} \approx 0.9 \text{ kPa}$.

6.4. In a standing wave, $\langle p^2 \rangle$ is maximum at pressure antinodes (where the velocity vanishes) and $\langle v^2 \rangle$ is maximum at pressure nodes (where the pressure vanishes). The pressure term $+f_1 \langle p^2 \rangle / (2\rho_0 c^2)$ adds positive energy at antinodes, pushing the particle away; the velocity term $-f_2 \frac{3\rho_0}{4} \langle v^2 \rangle$ lowers energy at nodes, pulling it in. The two statements are consistent because antinodes and nodes are $\lambda/4$ apart, so the two terms push toward the same point.

6.5. This is the standard momentum-flux derivation; the key steps are outlined in the problem statement, and the surface-integral identities give the coefficients V_p (monopole) and $-\frac{3\rho_0}{4} V_p$ (dipole) in (21).

6.6. The first term is the monopole (spherical, $\propto 1/r$, compressibility); the second is the dipole ($\propto \cos \theta / r^2$, density). Higher multipoles enter with $(ka)^{2\ell}$ prefactors, so for $ka \ll 1$ only $\ell = 0, 1$ survive.

6.7. With $U(z) = \frac{V_p P_0^2}{\rho_0 c^2} [f_1 \cos^2 kz - \frac{3}{4} f_2 \sin^2 kz]$, $F(z) = -\frac{dU}{dz} = \frac{V_p P_0^2 k}{\rho_0 c^2} (f_1 + \frac{3}{4} f_2) \sin kz \cos kz$ and at a node ($kz = \pi/2$): $k_t = \frac{d^2 U}{dz^2}|_{node} = \frac{2V_p P_0^2 k^2}{\rho_0 c^2} (f_1 + \frac{3}{4} f_2)$. Stability ($k_t > 0$) requires $f_1 + \frac{3}{4} f_2 > 0$. For polystyrene in air both $f_1, f_2 \approx 1$, so $k_t > 0$: the node is a genuine energy minimum.

6.8. (a) The phase $m\phi$ has gradient $\nabla(m\phi) = \frac{m}{r} \hat{\phi}$, an azimuthal momentum flux. The time-averaged angular momentum density $\propto r \times \langle \rho \mathbf{v} \rangle$ has z -component proportional to $m|A|^2$. (b) Off-axis, the azimuthal force $F_\phi = -\frac{1}{r} \frac{\partial U}{\partial \phi} \propto m$, giving $\tau_z = r F_\phi \propto m$: the particle spins about the singularity. (c) Summing (26) with $\varphi_n = m \arg(\mathbf{x}_t - \mathbf{x}_n)_\perp$ gives each transducer a phase that winds by m around the axis, so their coherent sum near the axis is locally $\propto e^{im\phi}$.

6.9. (a) At $\mathbf{x}_t$ each term in (26) has phase $k|\mathbf{x}_t - \mathbf{x}_n| + \varphi_n = 0$, so all add coherently. (b) U has the repulsive pressure term $+f_1 \langle p^2 \rangle / (2\rho_0 c^2)$; the $|P|$ maximum is a U maximum for $f_1 > 0$. A trap is a U minimum, i.e. a pressure node flanked by lobes. (c) Define a loss $\mathcal{L}(\varphi) = U(\mathbf{x}_t; \varphi)$ and iterate $\varphi_n \leftarrow \varphi_n - \eta \partial U(\mathbf{x}_t) / \partial \varphi_n$ (a projected gradient on the unit circle) until $U(\mathbf{x}_t)$ is minimized; the resulting hologram places a true potential well at $\mathbf{x}_t$.

6.10. For fixed acoustic power $P_0 \propto k$ (far-field amplitude grows with k at fixed surface velocity), so $U \propto P_0^2 \propto k^2 \propto \omega^2$ at fixed power. But $\lambda \propto 1/\omega$ shrinks, so the Rayleigh limit $ka < 1$ caps the particle size: higher frequency traps harder but only smaller objects.

6.11. (a) $k = 2\pi f/c = 2\pi \cdot 40 \times 10^3 / 343 \approx 733 \text{ m}^{-1}$, $ka = 733 \cdot 0.5 \times 10^{-3} \approx 0.37 < 1$. (b) $f_1 \approx 1 - \frac{1.2 \cdot 343^2}{1000 \cdot 1480^2} \approx 1.0$, $f_2 \approx 1.0$. (c) From Problem 6.3, $\Delta U \approx 5.9 \times 10^{-8} \text{ J}$; the stiffness $k_t \approx \frac{2V_p P_0^2 k^2}{\rho_0 c^2} \cdot \frac{7}{4} \approx 0.37 \text{ N/m}$, and for a $50 \mu\text{m}$ displacement the restoring force is $k_t \cdot 50 \mu\text{m} \approx 1.9 \times 10^{-5} \text{ N}$. (d) The weight is $mg \approx 5.1 \times 10^{-6} \text{ N}$, well below the maximum restoring force available; levitation is comfortably achieved at $P_0 = 1.5 \text{ kPa}$.

References

The primary source for the potential is L. P. Gor'kov, "On the forces acting on a small particle in an acoustical field in an ideal fluid," *Sov. Phys. Dokl.* **6**, 773 (1962). Standard treatments of the radiation stress and Rayleigh scattering appear in L. D. Landau and E. M. Lifshitz, *Fluid Mechanics* (Pergamon); the momentum-flux derivation follows H. Bruus, *Theoretical Microfluidics* (Oxford). The baffled-piston directivity is the Rayleigh–Sommerfeld result; holographic and vortex trap techniques are described in A. Marzo, S. A. Seah, B. W. Drinkwater et al., "Holographic acoustic elements for manipulation of levitated objects," *Nat. Commun.* **6**, 8661 (2015).